

OT ATTACK AND DETECTION

LABORATORY REPORT

Water Treatment Plant PLC Simulation

OpenPLC | Kali Linux | Suricata IDS | Wazuh SIEM

Prepared by

Sheheryar Altaf

OT Security Researcher

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1. EXECUTIVE SUMMARY

This report documents a complete Operational Technology (OT) security assessment performed in a controlled laboratory environment. A water treatment plant was simulated using the OpenPLC Runtime platform. An unauthorized Modbus TCP write attack was launched from a Kali Linux machine to manipulate PLC memory registers without any authentication.

The attack successfully changed sensor register values, forcing the pump and alarm outputs into an active state. This demonstrated the zero authentication vulnerability present in the Modbus TCP protocol.

Network traffic was captured using Wireshark. Custom Suricata IDS rules were developed to detect unauthorized Modbus write commands. All alerts were forwarded to the Wazuh SIEM platform and mapped to MITRE ATT&CK; for ICS techniques: Remote System Discovery (T0846), Unauthorized Command Message (T1692.001), Manipulation of Control (T0831), and Loss of Availability (T0826).

2. LABORATORY ENVIRONMENT

The lab was built using VMware Workstation with two virtual machines connected through a NAT network. The Ubuntu VM (192.168.71.128) hosted OpenPLC Runtime on port 502, Suricata IDS, and the full Wazuh SIEM stack. The Kali VM (192.168.71.129) was configured as the attacker machine.

Component	Details
Ubuntu VM	192.168.71.128 (OpenPLC, Suricata, Wazuh)
Kali VM	192.168.71.129 (Attacker)
Network	VMware NAT (vmnet8)
Protocol	Modbus TCP, Port 502

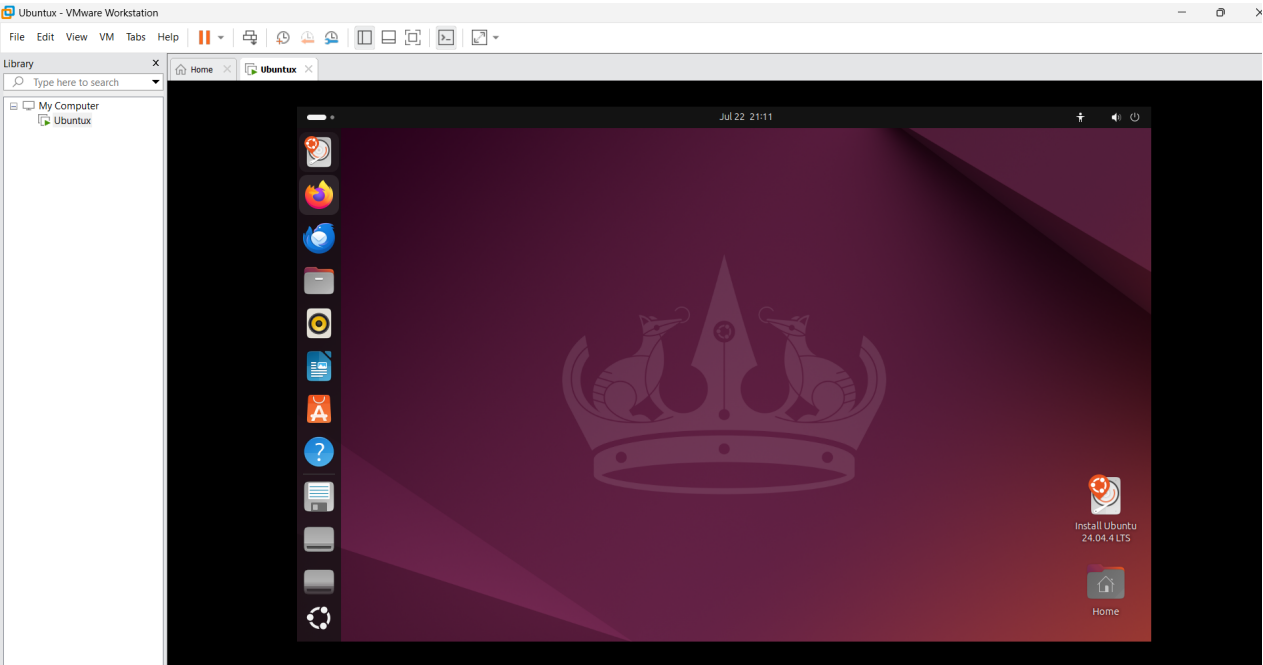


Figure 1: VMware Workstation with Ubuntu VM Running

3. PLC PROGRAM DESIGN

3.1 Ladder Logic

A water treatment plant scenario was designed with three sensor inputs (low_level, high_level, alarm_level) mapped to %MW0 through %MW2, and two control outputs (pump, alarm) mapped to %QX0.0 and %QX0.1. The pump activates when low_level exceeds zero. The alarm triggers when alarm_level is non zero.

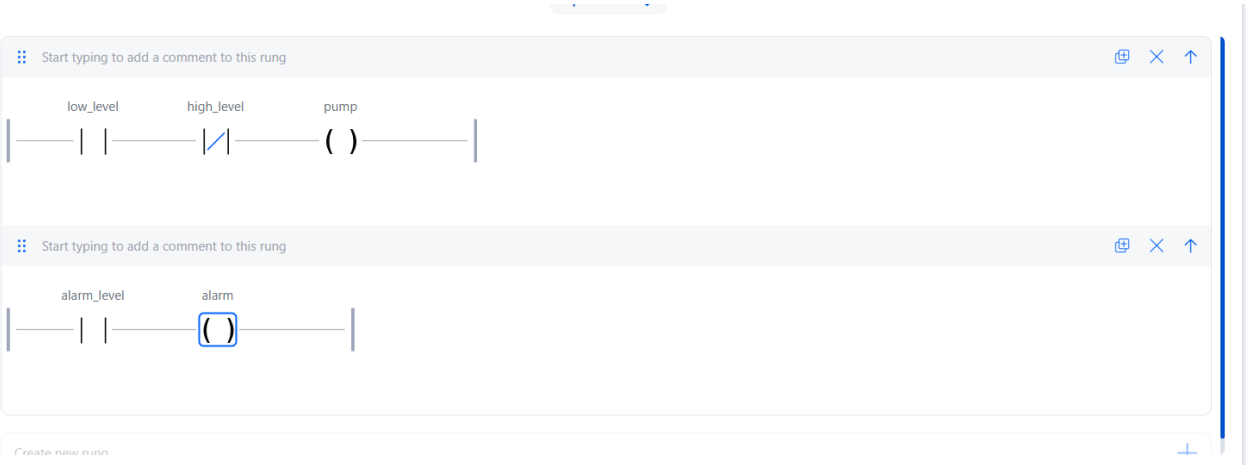


Figure 2: Ladder Logic Design for Water Tank Control Program

3.2 Variable Mapping

Variable	Type	PLC Address	Modbus Address	Role
low_level	WORD	%MW0	1024	Tank bottom sensor
high_level	WORD	%MW1	1025	Tank 80% sensor
alarm_level	WORD	%MW2	1026	Overflow sensor
pump	BOOL	%QX0.0	Coil 0	Inlet valve
alarm	BOOL	%QX0.1	Coil 1	Buzzer/Alert

3.3 Compilation and Deployment

The Structured Text program was uploaded to OpenPLC and compiled successfully. The runtime was started on port 502, making the PLC accessible over Modbus TCP.

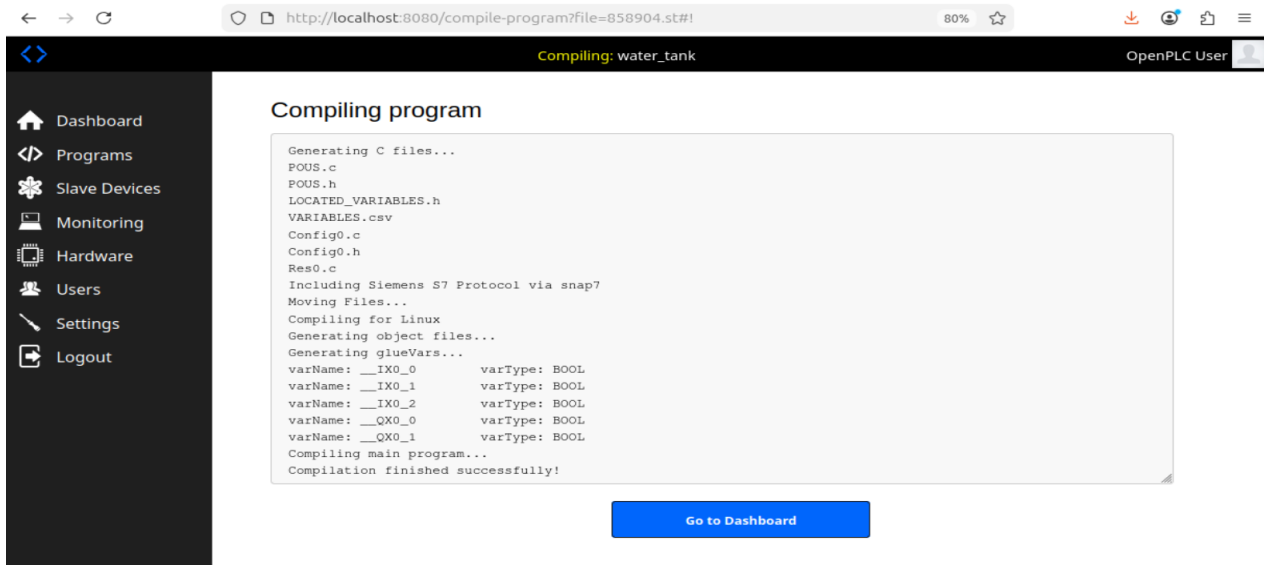


Figure 3: Successful Compilation of PLC Program

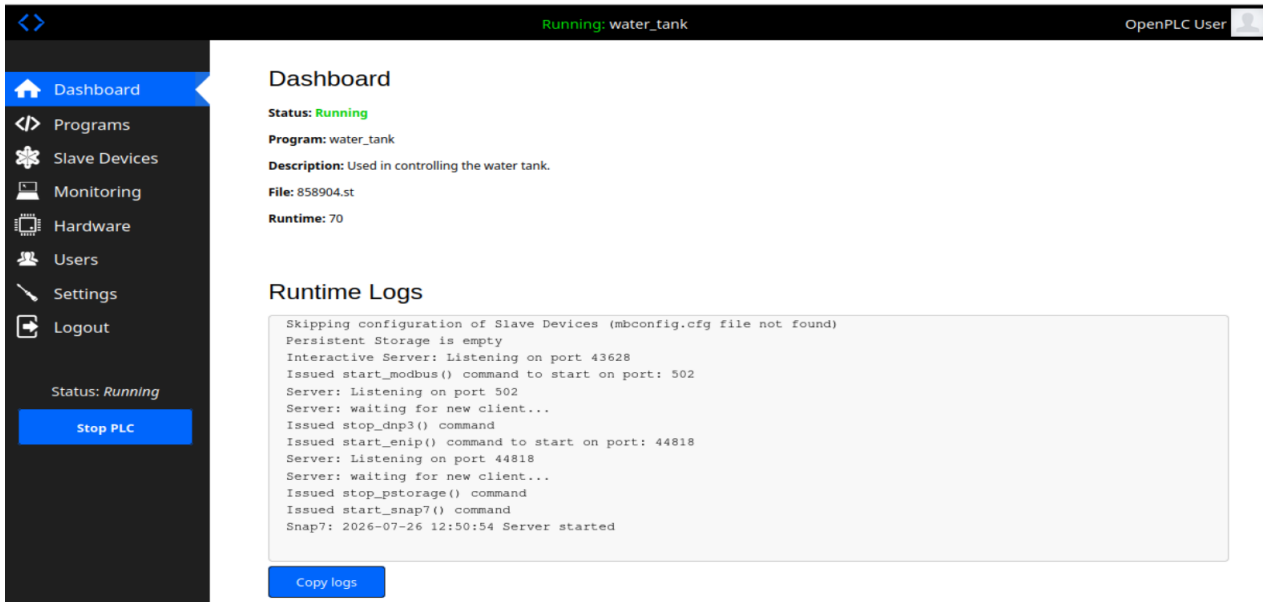


Figure 4: OpenPLC Runtime Active on Port 502 (Status: Running)

3.4 Modbus Address Mapping

OpenPLC maps %MW variables starting at Modbus address 1024, not 0. This was confirmed by inspecting the LOCATED_VARIABLES.h file generated during compilation.

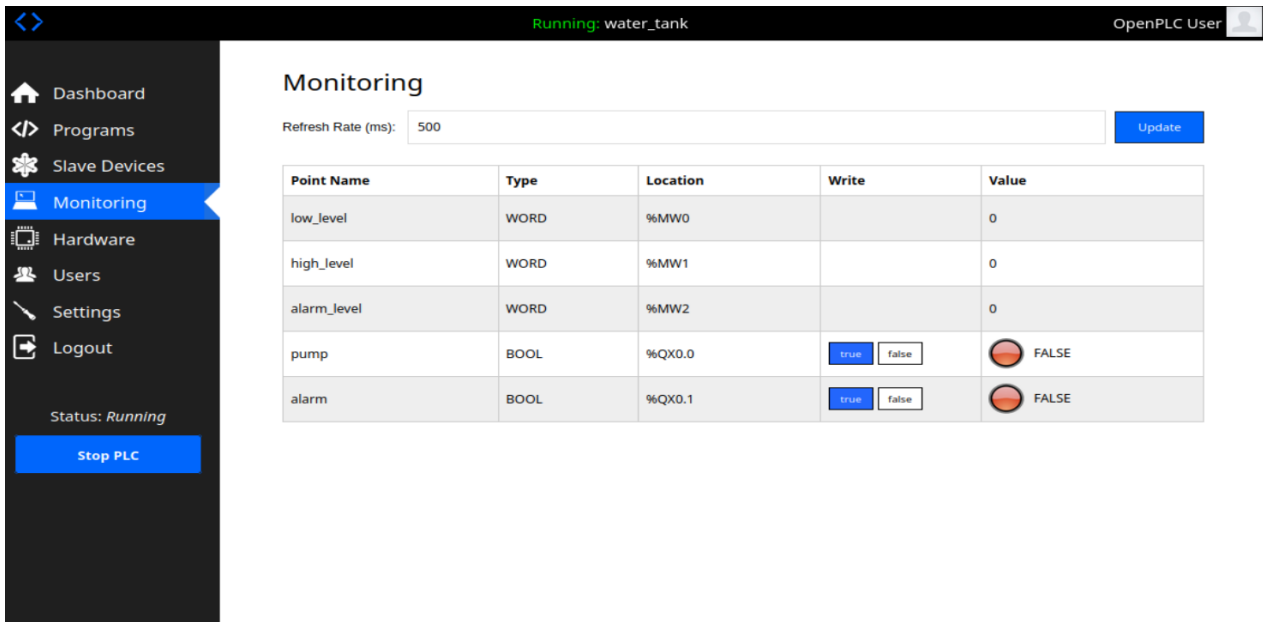


Figure 5: LOCATED_VARIABLES.h Confirming MW0 = Address 1024

4. ATTACK SIMULATION

4.1 Baseline State

Before the attack, the monitoring page was inspected. All sensor values were zero. Both pump and alarm coils were FALSE.



The screenshot displays the 'Monitoring' page of a PLC interface. The top status bar indicates 'Running: water_tank' and the user 'OpenPLC User'. A left sidebar contains navigation links: Dashboard, Programs, Slave Devices, Monitoring (selected), Hardware, Users, Settings, and Logout. Below the sidebar, the status is 'Running' with a 'Stop PLC' button. The main area shows a 'Monitoring' section with a 'Refresh Rate (ms): 500' input and an 'Update' button. A table lists five points: low_level, high_level, alarm_level, pump, and alarm. The first three are WORD types with values of 0. The last two are BOOL types with values of FALSE. The 'Write' column for the BOOL points includes 'true' and 'false' buttons.

Point Name	Type	Location	Write	Value
low_level	WORD	%MW0		0
high_level	WORD	%MW1		0
alarm_level	WORD	%MW2		0
pump	BOOL	%QX0.0	true false	☐ FALSE
alarm	BOOL	%QX0.1	true false	☐ FALSE

Figure 6: Monitoring Page Showing Baseline (All Values Zero, Coils FALSE)

4.2 Attack Execution

A Python script using pymodbus was executed from Kali Linux. It read addresses 1024 to 1026, confirmed baseline [0, 0, 0], then sent two FC6 (Write Single Register) commands. Address 1024 was set to 1 and address 1026 was set to 1. No authentication was required. The final read confirmed the state [1, 0, 1].

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ python3 attack.py  
Before attack: [0, 0, 0]  
After attack: [1, 0, 1]  
  
(kali@kali)-[~]  
$ python3 attack.py  
Before attack: [1, 0, 1]  
After attack: [1, 0, 1]  
  
(kali@kali)-[~]  
$
```

Figure 7: Attack Script Output: Before [0,0,0] and After [1,0,1]

4.3 Attack Confirmation

The monitoring page confirmed the attack. The pump coil switched to TRUE and the alarm coil switched to TRUE. The PLC was hijacked without any credentials.

Monitoring

Refresh Rate (ms): Update

Point Name	Type	Location	Write	Value
low_level	WORD	%MW0		1
high_level	WORD	%MW1		0
alarm_level	WORD	%MW2		1
pump	BOOL	%QX0.0	☒ true ☐ false	 TRUE
alarm	BOOL	%QX0.1	☒ true ☐ false	 TRUE

Figure 8: Monitoring After Attack (low_level=1, alarm_level=1, Pump=TRUE, Alarm=TRUE)

5. NETWORK TRAFFIC ANALYSIS

The complete attack was captured using Wireshark on the ens33 interface with filter tcp.port == 502.

Packet(s)	Description
223 to 225	TCP three way handshake (SYN, SYN ACK, ACK)
226	Modbus Read Holding Registers (pre attack check)
232	Modbus Read response [0, 0, 0]
234	Modbus Write FC6 to address 1024 (ATTACK 1)
237	Modbus Write FC6 to address 1026 (ATTACK 2)
239	Modbus Read Holding Registers (verification)
240	Modbus Read response [1, 0, 1] (confirmed)
241 to 243	TCP FIN ACK (connection closed)

No.	Time	Source	Destination	Protocol	Length	Info
223	31.310723975	192.168.35.129	192.168.35.128	TCP	74	37086 → 502 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 S
224	31.317222194	192.168.35.128	192.168.35.129	TCP	74	502 → 37086 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0
225	31.330077335	192.168.35.129	192.168.35.128	TCP	66	37086 → 502 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSva
226	31.350114503	192.168.35.129	192.168.35.128	Modbus...	78	Query: Trans: 1; Unit: 1, Func: 3: Read
227	31.350131991	192.168.35.128	192.168.35.129	TCP	66	502 → 37086 [ACK] Seq=1 Ack=13 Win=65280 Len=0 TSv
232	33.378798119	192.168.35.128	192.168.35.129	Modbus...	81	Response: Trans: 1; Unit: 1, Func: 3: Read
233	33.388786196	192.168.35.129	192.168.35.128	TCP	66	37086 → 502 [ACK] Seq=13 Ack=16 Win=64256 Len=0 TS
234	33.394867793	192.168.35.129	192.168.35.128	Modbus...	78	Query: Trans: 2; Unit: 1, Func: 6: Writ
235	33.395163927	192.168.35.128	192.168.35.129	TCP	66	502 → 37086 [ACK] Seq=16 Ack=25 Win=65280 Len=0 TS
236	33.400008951	192.168.35.128	192.168.35.129	Modbus...	78	Response: Trans: 2; Unit: 1, Func: 6: Writ
237	33.401036923	192.168.35.129	192.168.35.128	Modbus...	78	Query: Trans: 3; Unit: 1, Func: 6: Writ
238	33.401357694	192.168.35.128	192.168.35.129	Modbus...	78	Response: Trans: 3; Unit: 1, Func: 6: Writ
239	33.411045101	192.168.35.129	192.168.35.128	Modbus...	78	Query: Trans: 4; Unit: 1, Func: 3: Read
240	33.411216300	192.168.35.128	192.168.35.129	Modbus...	81	Response: Trans: 4; Unit: 1, Func: 3: Read
241	33.413130574	192.168.35.129	192.168.35.128	TCP	66	37086 → 502 [FIN, ACK] Seq=49 Ack=55 Win=64256 Len
242	33.414151774	192.168.35.128	192.168.35.129	TCP	66	502 → 37086 [FIN, ACK] Seq=55 Ack=50 Win=65280 Len

Figure 9: Wireshark Packet List Showing Complete Attack Sequence



6. SURICATA IDS DETECTION

Three custom Suricata rules were deployed. A pass rule whitelists local PLC traffic. Two alert rules detect Function Code 6 and Function Code 16 using content match at offset 7 with depth 1. The function code is exactly one byte, not two. Using depth:2 with |00 06| failed. The corrected pattern |06| with depth:1 works.

```
ubuntu@ubuntu-VMware-Virtual-Platform:~$ cat /var/log/suricata/fast.log
07/29/2026-14:04:35.401783  [**] [1:1000002:1] Modbus Port Traffic Detected [**] [Classification: (null)] [Priority: 3]
{TCP} 192.168.71.129:51524 -> 192.168.71.128:502
07/29/2026-15:32:01.195337  [**] [1:1000001:1] Modbus Traffic Detected [**] [Classification: (null)] [Priority: 3] {TCP}
192.168.71.129:33180 -> 192.168.71.128:502
07/29/2026-16:05:57.878075  [**] [1:1000001:2] Unauthorized Modbus Write FC6 [**] [Classification: (null)] [Priority: 3]
{TCP} 192.168.71.129:57114 -> 192.168.71.128:502
07/29/2026-16:05:57.879793  [**] [1:1000001:2] Unauthorized Modbus Write FC6 [**] [Classification: (null)] [Priority: 3]
{TCP} 192.168.71.129:57114 -> 192.168.71.128:502
ubuntu@ubuntu-VMware-Virtual-Platform:~$
```

Figure 11: Suricata fast.log Showing FC6 Alerts from Kali (192.168.71.129)

7. WAZUH SIEM INTEGRATION

7.1 Dashboard Overview

Suricata alerts are forwarded to Wazuh through the eve.json log file. The Wazuh dashboard provides centralized visibility into all security events with severity classification.

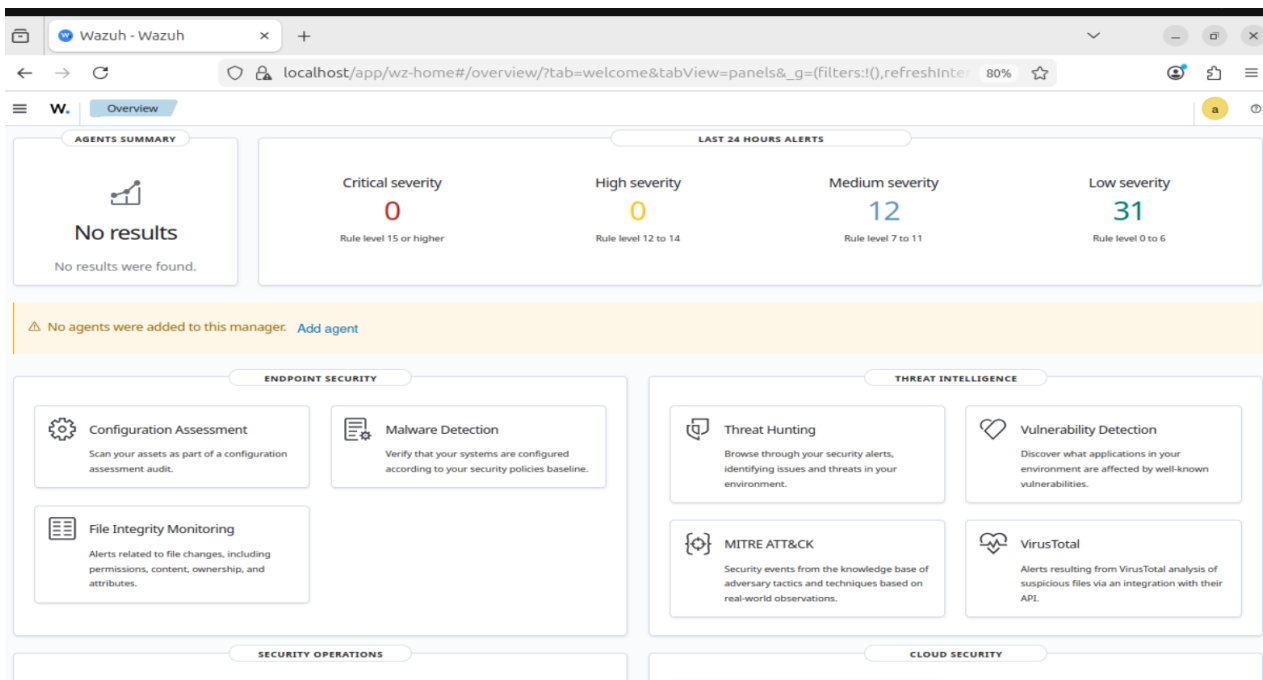


Figure 12: Wazuh Overview Dashboard with Security Event Summary

7.2 Custom Rule with MITRE Mapping

A custom Wazuh rule (ID 100050, level 13) was created in local_rules.xml. It triggers as a child of built in rule 86601. Three MITRE ATT&CK; for ICS technique IDs were embedded: T0846, T0831, and T0826.

OT Attack: Unauthorized Modbus Write on PLC [View alerts of this Rule](#)

Information

ID 100001	Level 13	File modbus_wazuh.xml	Path etc/rules
---------------------	--------------------	---	--------------------------

Groups
Modbus, suricata

Details

If_sid 86601	Alert.Signature pattern: Unauthorized Modbus Write
------------------------	--

Compliance

Related rules

ID ↑	Description	Groups	Compliance	Level	File
10000 1	OT Attack: Unauthorized Modbus Write on PLC	Modbus, suricata	MITRE	13	modbus_wazuh.xml

Rows per page: 10

Figure 13: Wazuh Custom Rule with MITRE ATT&CK; Badge (Level 13)

7.3 Alert Verification

The Wazuh Discover view shows detailed alert records. Each alert includes source IP, signature ID, severity, and timestamp.

Columns		Sort fields	
Time (timestamp)	Source		
Jul 29, 2026 @ 16:05:57.986	input.type log agent.name ubuntu-VMware-Virtual-Platform agent.id 000 managername ubuntu-VMware-Virtual-Platform data.capture_file /var/log/suricata/log.pcap.1785323143 data.app_proto modbus data.in_iface ens33 data.src_ip 192.168.71.129 data.src_port 57114 data.event_type alert data.alert.severity 3 data.alert.signature_id 1000001 data.alert.rev 2 data.alert.gid 1 data.alert.signature Unauthorized Modbus Write FC6 data.alert.action allowed data.modbus.request.transaction_id 1 data.modbus.request.access_t...		
Jul 29, 2026 @ 16:05:57.986	input.type log agent.name ubuntu-VMware-Virtual-Platform agent.id 000 managername ubuntu-VMware-Virtual-Platform data.capture_file /var/log/suricata/log.pcap.1785323143 data.app_proto modbus data.in_iface ens33 data.src_ip 192.168.71.129 data.src_port 57114 data.event_type alert data.alert.severity 3 data.alert.signature_id 1000001 data.alert.rev 2 data.alert.gid 1 data.alert.signature Unauthorized Modbus Write FC6 data.alert.action allowed data.flow_id 1663686628560902.000000 data.dest_ip 192.168.71.12...		

Figure 14: Wazuh Alert Detail for Modbus Attack Events

Columns		Sort fields	
Time (timestamp)	Source		
Jul 29, 2026 @ 16:05:57.986	input.type log agent.name ubuntu-VMware-Virtual-Platform agent.id 000 managername ubuntu-VMware-Virtual-Platform data.capture_file /var/log/suricata/log.pcap.1785323143 data.app_proto modbus data.in_iface ens33 data.src_ip 192.168.71.129 data.src_port 57114 data.event_type alert data.alert.severity 3 data.alert.signature_id 1000001 data.alert.rev 2 data.alert.gid 1 data.alert.signature Unauthorized...		
Jul 29, 2026 @ 16:05:57.986	input.type log agent.name ubuntu-VMware-Virtual-Platform agent.id 000 managername ubuntu-VMware-Virtual-Platform data.capture_file /var/log/suricata/log.pcap.1785323143 data.app_proto modbus data.in_iface ens33 data.src_ip 192.168.71.129 data.src_port 57114 data.event_type alert data.alert.severity 3 data.alert.signature_id 1000001 data.alert.rev 2 data.alert.gid 1 data.alert.signature Unauthorized...		
Jul 29, 2026 @ 15:32:02.500	input.type log agent.name ubuntu-VMware-Virtual-Platform agent.id 000 managername ubuntu-VMware-Virtual-Platform data.in_iface ens33 data.src_ip 192.168.71.129 data.src_port 33180 data.event_type alert data.alert.severity 3 data.alert.signature_id 1000001 data.alert.rev 1 data.alert.gid 1 data.alert.signature Modbus Traffic Detected data.alert.action allowed data.flow_id 557495280189287.000000 dat...		

Figure 15: Wazuh Dashboard Showing All OT Attack Alerts

8. MITRE ATT&CK; FOR ICS MAPPING

The observed attack was mapped to four MITRE ATT&CK; for ICS techniques.

Tactic	ID	Technique	Evidence
Discovery	T0846	Remote System Discovery	PLC at 192.168.71.128:502
Evasion	T1692.001	Unauthorized Command Message	FC6 write without authentication
Impact	T0831	Manipulation of Control	MW0, MW2 modified, pump forced ON
Impact	T0826	Loss of Availability	Pump forced ON, overflow scenario

9. KEY FINDINGS AND RECOMMENDATIONS

9.1 Findings

1. Modbus TCP provides zero authentication. Any network device can write to PLC registers without credentials.
2. OpenPLC maps %MW variables to Modbus address 1024 and above. This vendor specific behavior must be understood before security testing.
3. Suricata detects Modbus attacks by matching function code bytes at offset 7 with depth 1. Incorrect depth causes missed detections.
4. Wazuh SIEM integrates with Suricata through eve.json. Custom rules with MITRE ICS mappings provide centralized threat classification.
5. Pass rules in Suricata whitelist authorized traffic while alerting on unauthorized sources.

9.2 Recommendations

Network segmentation should isolate OT networks from IT networks. Access to Modbus devices should be restricted through OT firewalls or protocol aware gateways.

Continuous monitoring through Wazuh with MITRE ATT&CK; for ICS mapped rules should be maintained. SOC teams should be trained on OT alert investigation and response.

9.3 Tools Used

Tool	Purpose
OpenPLC Runtime	PLC simulation for water tank control
Kali Linux	Attack platform with pymodbus
Wireshark	Network traffic capture and analysis
Suricata	IDS for Modbus function code detection
Wazuh	SIEM with MITRE ATT&CK mapping
VMware Workstation	Lab virtualization with NAT isolation

End of Report